LONG-TERM SAFETY OF ADJUNCTIVE CENOBAMATE IN PATIENTS WITH UNCONTROLLED FOCAL SEIZURES: OPEN-LABEL EXTENSION OF A RANDOMIZED CLINICAL STUDY

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BACKGROUND

- Cenobamate (CNB) is an antiepileptic drug (AED) being investigated for adults with focal epilepsy
- Results from two clinical studies demonstrated substantial and statistically significant reductions in seizure frequency and high responder rates, including seizure freedom, in patients with uncontrolled focal seizures treated with adjunctive CNB[1,2]
- In the 12-week, double-blind (DB), placebo (PBO)-controlled, phase 2, YKP3089C013 (C013) study, the median percentage seizure frequency reduction/28 days was 55.6% for CNB 200 mg compared with 21.5% for PBO (*P*<0.0001), and 28.3% of CNB-treated patients were seizure-free during the 6-week maintenance treatment phase compared with 8.8% for placebo (*P*=0.0001)[2]
- During the 12-week DB study, the most frequently (>10%) reported treatment-emergent adverse events (TEAEs) for cenobamate were somnolence, dizziness, headache, nausea, and fatigue[2]
- Patients who completed the 12-week DB treatment period were eligible to continue in an open-label extension (OLE) on CNB
- As seizure diaries were not collected in the C013 OLE study, we report here the long-term safety and retention outcomes from the ongoing OLE of the C013 study, which are often used as an indicator of overall treatment satisfaction (incorporating both efficacy and safety)

METHODS

Study Design

- Patients in the multicenter, multinational, DB, randomized, PBO-controlled C013 study (ClinicalTrials.gov NCT01397968) were randomized to receive PBO or CNB at a target dose of 200 mg once daily for 12 weeks (6-week titration phase, 6-week maintenance phase)
- Key inclusion criteria[2]:
 - Adult patients aged 18-65 years old with a diagnosis of treatment-resistant focal epilepsy[3,4]
 - History of epilepsy for at least 2 years
 - Taking 1-3 AEDs at stable doses for at least 12 weeks before randomization
 - Within the 8-week baseline period, had ≥3 focal aware seizures with motor component, including aphasia and other observable symptoms; focal impaired awareness; or focal to bilateral tonic-clonic (secondarily generalized) seizures per month; and no consecutive 21-day seizure-free period
- Key exclusion criteria[2]:
 - Receiving phenytoin, phenobarbital, or metabolites of these drugs due to potential for drug-drug interactions
 - Receiving vigabatrin within the past year, felbamate for less than 18 continuous months, or intermittent rescue benzodiazepines more than once a month within the past month
 - Status epilepticus
 - Alcoholism, drug abuse, or drug addiction
- In the DB study, flexible dose adjustment was based on tolerability
 - Patients who tolerated each dose for 2 weeks were increased to the next higher dose. If a patient could not tolerate the next higher dose, he or she was continued on the current dose or had their dose reduced.
- At the end of the DB study, patients were either tapered off study drug for one week before undergoing transition to open-label treatment or underwent a blinded conversion to open-label CNB without being tapered off, if medically warranted (following protocol amendment)
- All patients entering the OLE received CNB at 100 mg/day with subsequent dose increases of 50 mg every 2 weeks as tolerated; the maximum allowed daily dose of CNB during the OLE was 400 mg

Analysis

- Data were summarized with descriptive statistics
- The OLE safety population was defined as all patients who entered the OLE and took at least 1 dose of CNB

RESULTS

Patients

- Of the 201 patients who completed the C013 DB study, 43 were from study sites that did not participate in the OLE
- Of the remaining 158 patients, 149 entered the OLE, including 76 patients originally randomized to CNB (CNB/CNB) and 73 patients originally randomized to PBO who transitioned to CNB (PBO/CNB)
- Baseline demographics were similar between groups (**Table 1**); 89% (n=132) of patients were taking 2 or >2 baseline AEDs

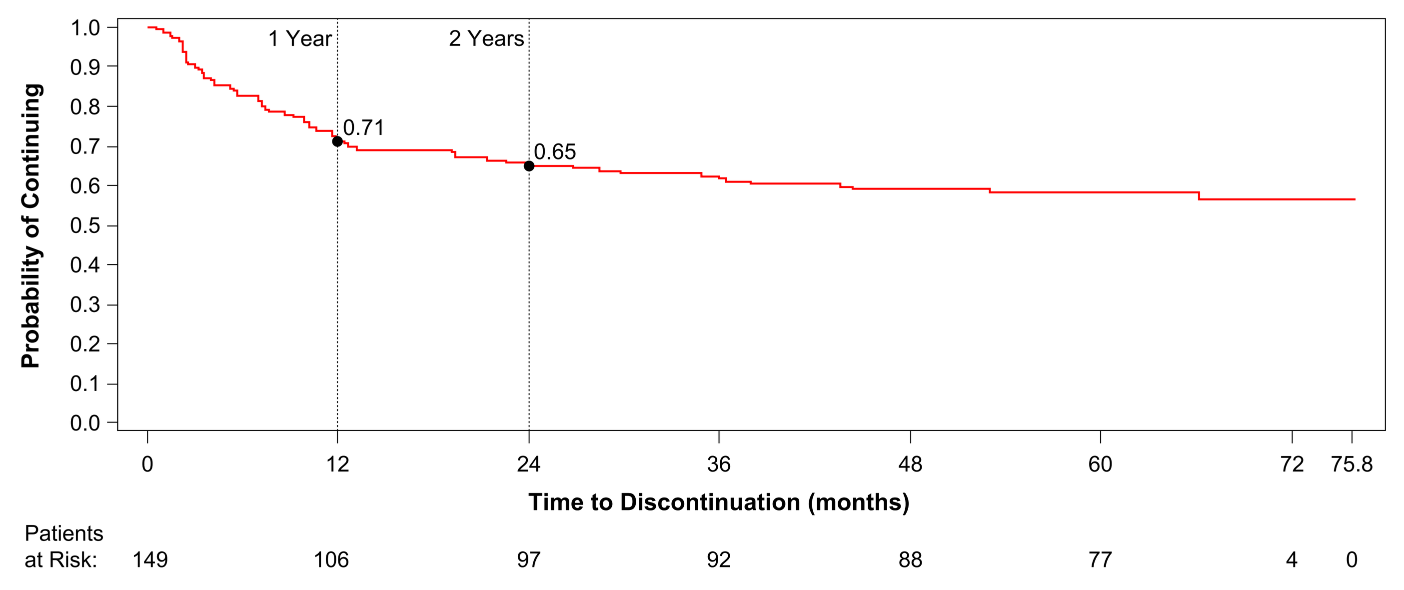
Table 1. Demographics and Baseline Characteristics (OLE Population)

	All CNB (n=149)	CNB/CNB (n=76)	PBO/CNB (n=73)
Age, years, mean (SD)	37.6 (10.9)	36.9 (11.1)	38.3 (10.7)
Female, n (%)	77 (51.7)	42 (55.3)	35 (47.9)
Race, n (%)			
Caucasian/White	99 (66.4)	50 (65.8)	49 (67.1)
Asian	37 (24.8)	19 (25.0)	18 (24.7)
African American/Black	5 (3.4)	3 (3.9)	2 (2.7)
Other/Unknown	8 (5.4)	4 (5.3)	4 (5.5)
Baseline[a] AEDs, n (%)			
1	17 (11.4)	13 (17.1)	4 (5.5)
2	69 (46.3)	34 (44.7)	35 (47.9)
>2	63 (42.3)	29 (38.2)	34 (46.6)

[a]Baseline AEDs defined as AEDs starting prior to and ongoing at the time of first dose in double-blind period.
AED, anti-epileptic drug; CNB, cenobamate; OLE, open-label extension; PBO, placebo; SD, standard deviation.

- As of April 2018, 86 patients (57.7%) remained in the OLE, including 52.6% (40/76) of CNB/CNB patients and 63.0% (46/73) of PBO/CNB patients. The median duration of CNB exposure was 60.6 months (range, 0.3-79.4 months)
- After 1 year of treatment, 71% of patients remained in the OLE; after 2 years, 65% remained in the OLE (**Figure 1**)
- Among the patients who continued treatment at 12 months, 81% were still receiving CNB after a median duration of 60 months

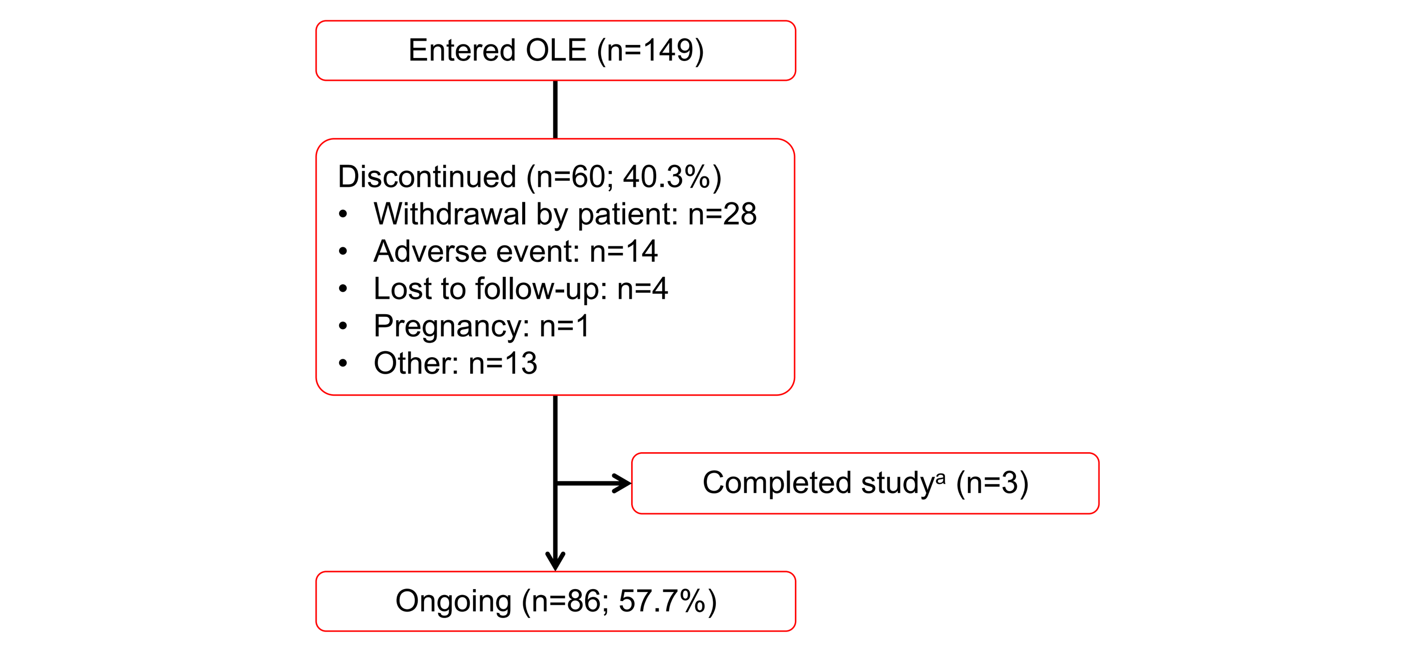
Figure 1. Kaplan-Meier Plot of Continuation During the OLE (Safety Population)


1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Probability of Continuing
1 Year
2 Years
0.71
0.65
0
12
24
36
48
60
72
75.8
Time to Discontinuation (months)
Patients at Risk:
149
106
97
92
88
77
4
0

Event = early discontinuation from OLE. Patients who completed the study and patients ongoing at date of data cut-off are considered as censored.
OLE, open-label extension.

- Reasons for discontinuation overall are shown in **Figure 2**

Figure 2. Patient Disposition and Reason for Discontinuation (OLE Population)


Entered OLE (n=149)
Discontinued (n=60; 40.3%)
• Withdrawal by patient: n=28
• Adverse event: n=14
• Lost to follow-up: n=4
• Pregnancy: n=1
• Other: n=13
Completed study[a] (n=3)
Ongoing (n=86; 57.7%)

[a]As reported on the end of study case report form.

OLE, open-label extension.

Dose

- The median modal daily CNB dose was 200 mg (range, 50-400 mg)
- The last recorded dose received among patients at or before the data cut-off was 400 mg/day (8.1%, 12/149), 350 mg/day (4.0%, 6/149), 300 mg/day (6.7%, 10/149), 250 mg/day (5.4%, 8/149), 200 mg/day (26.2%, 39/149); 150 mg/day (8.1%, 12/149); 100 mg/day (22.8%, 34/149); and ≤50 mg/day (18.8%, 28/149)

Safety

- TEAEs occurred in 88.6% (n=132) of patients during the OLE (safety population, N=149)
- Dizziness, headache, and somnolence were the most frequently reported TEAEs (**Table 2**)
- Serious TEAEs occurred in 22.8% (n=34) of patients; serious TEAEs occurring in >1% of patients were seizure (n=5) and pneumonia, sepsis, and osteoarthritis (n=2 each)
 - One sudden unexplained death in epilepsy (SUDEP) occurred during the OLE
- TEAEs leading to discontinuation were reported in 9.4% (n=14) of patients overall, including 9 CNB/CNB patients and 5 PBO/CNB patients
 - TEAEs leading to discontinuation in >1% of patients were fatigue (n=2) and ataxia (n=2)

Table 2. Summary of TEAEs

n, %	All CNB (N=149)	CNB/CNB (n=76)	PBO/CNB (n=73)
Any TEAE	132 (88.6)	66 (86.8)	66 (90.4)
Serious TEAEs	34 (22.8)	22 (28.9)	12 (16.4)
TEAEs ≥10% in any group			
Dizziness	48 (32.2)	24 (31.6)	24 (32.9)
Headache	39 (26.2)	20 (26.3)	19 (26.0)
Somnolence	31 (20.8)	14 (18.4)	17 (23.3)
Viral upper respiratory tract infection	26 (17.4)	15 (19.7)	11 (15.1)
Upper respiratory tract infection	20 (13.4)	10 (13.2)	10 (13.7)
Nausea	16 (10.7)	5 (6.6)	11 (15.1)
Fatigue	16 (10.7)	6 (7.9)	10 (13.7)
Urinary tract infection	15 (10.1)	8 (10.5)	7 (9.6)
Diplopia	14 (9.4)	5 (6.6)	9 (12.3)
Back pain	12 (8.1)	8 (10.5)	4 (5.5)
Pain in extremity	12 (8.1)	6 (7.9)	6 (8.2)
Contusion	11 (7.4)	8 (10.5)	3 (4.1)

TEAEs, treatment-emergent adverse events.

- In terms of timing for the most common TEAEs, the median onset for both dizziness and somnolence occurred early in the OLE for the PBO/CNB group (17 and 16 days, respectively) and later in the OLE for the CNB/CNB group (56 and 102.5 days, respectively)
 - A similar trend, however, was not noted for headache

CONCLUSIONS

- Adjunctive CNB was generally well tolerated by those patients treated over the longer term (up to 79 months), and 81% of patients who received treatment at 12 months were still receiving CNB after a median duration of 60 months
- The 1-year (71%) and 2-year (65%) retention rates with CNB are similar to the retention rates reported for other recently approved adjunctive AEDs for focal seizures[5]
- Most TEAEs were central nervous system-related and consistent with the double-blind study[2]

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DISCLOSURES

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